

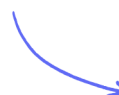
Structured Questions

Amount of Substance

Concentration Calculations / Reacting Mass Calculations / Reacting Volume Calculations / Calculations of Product

Medium (2 questions)	/19
Hard (1 question)	/8
Total Marks	/27

Scan here to return to the course
or visit [savemyexams.com](https://www.savemyexams.com)



Medium Questions

1 (a) This question is about silicon and carbon.

Silicon is a semiconductor.

i) Data obtained using the mass spectrum of silicon are shown.

Isotope mass number	Relative abundance
28	92.17
29	4.71
30	3.12

Calculate the relative atomic mass of silicon to **two** decimal places.

(2)

ii) Suggest a reason why there is a small peak in the mass spectrum of silicon at $m/z = 14$

(1)

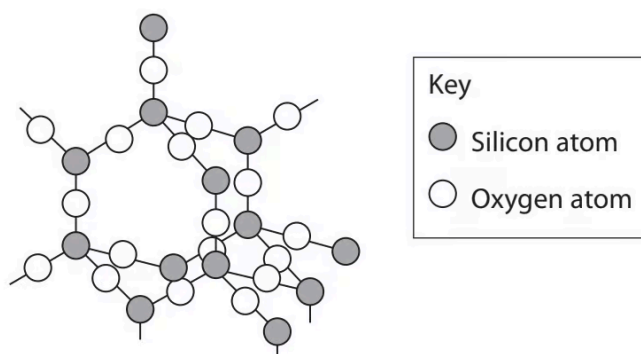
iii) Complete the table to show the number of protons and neutrons in each isotope of silicon.

Isotope	Number of protons	Number of neutrons
^{28}Si		
^{29}Si		
^{30}Si		

(1)

(4 marks)

- (b) Silicon dioxide, SiO_2 , is the main constituent of sand and has a giant lattice structure similar to that of diamond.



Crystalline silicon dioxide is used on the surface of semiconductor devices to provide a heat-resistant, electrically insulating layer.

Explain how the structure and bonding of silicon dioxide make it useful for this application.

.....
.....
.....
(3 marks)

- (c) Calcium silicate is formed in the removal of silicon dioxide impurities in the extraction of iron from its ores. A sample of calcium silicate composed of calcium, silicon and oxygen was found to contain 12.0 g of calcium, 8.43 g of silicon and 14.47 g of oxygen.

Determine the empirical formula of calcium silicate. You **must** show your working.

.....
.....
.....
(3 marks)

- (d) Carbon dioxide is a gas at room temperature. A fizzy drink is canned at 5.0°C and 1.3×10^5 Pa and contains approximately 3 g of carbon dioxide.

Calculate the volume, in cm^3 , occupied by 3.00 g of carbon dioxide gas at 5.0°C and 1.3×10^5 Pa.

$$[pV = nRT \quad R = 8.31 \text{ J mol}^{-1} \text{ K}^{-1}]$$

.....
.....
.....
(4 marks)

- 2 (a)** A 0.210 g sample of a volatile organic liquid **C** is injected into a gas syringe and heated in an oven. At 100 kPa and 358 K, the syringe contains 72.5 cm³ of gas.

Calculate the molar mass of **C**.

$$[pV = nRT \quad R = 8.31 \text{ J mol}^{-1} \text{ K}^{-1}]$$

(4 marks)

- (b)** The organic liquid **C** is a hydrocarbon. Give a possible name or formula for **C**, using your answer in (a).

(1 mark)

Hard Questions

- 1 (a) The hydride of arsenic, arsine, is a toxic gas used in the production of semiconductors.

Draw a dot-and-cross diagram for arsine, AsH_3 .

(1 mark)

- (b) Arsine is a reducing agent and reacts with cerium(IV) sulfate solution, forming arsenic.

The data from an experiment are shown.

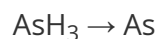
Volume of arsine gas = 350 cm^3 at 115000 Pa and 20°C

Volume of cerium(IV) sulfate solution = 488 cm^3

Concentration of cerium(IV) sulfate solution = $0.102 \text{ mol dm}^{-3}$

- i) Complete the half-equation.

(1)



- ii) Calculate the final oxidation state of the cerium ion formed in the reaction.

(6)

(7 marks)